EE2001 Electrical Circuits- I:

Continuation of the Topic:
Sinosoidal Steady-State Analysis

08 January 2024

Series, Parallel and Delta-to-Wye Simplifications

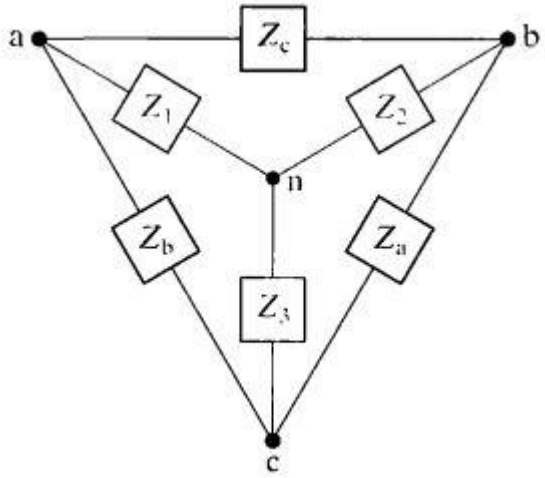
Delta-to-Wye Transformations

- The Δ-to-Y transformation for resistive circuits also applies to impedances.
- The Y impedances as functions of Δ impedances:

$$Z_1 = \frac{Z_b Z_c}{Z_a + Z_b + Z_c}$$

$$Z_2 = \frac{Z_c Z_a}{Z_a + Z_b + Z_c}$$

$$Z_3 = \frac{Z_a Z_b}{Z_a + Z_b + Z_c}$$



Series, Parallel and Delta-to-Wye Simplifications

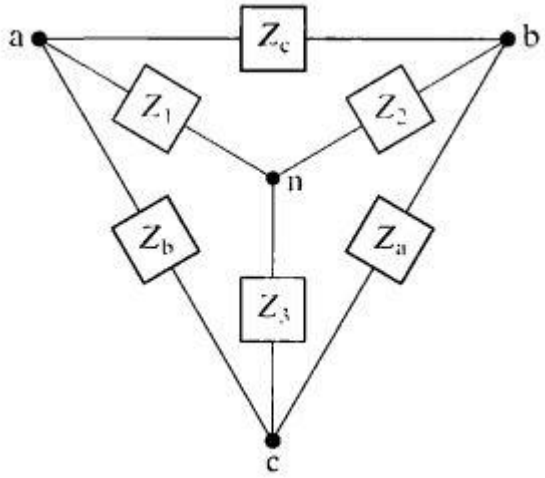
Delta-to-Wye Transformations

- The Δ impedances as functions of Y impedances:

$$Z_a = \frac{Z_1Z_2 + Z_2Z_3 + Z_3Z_1}{Z_1}$$

$$Z_b = \frac{Z_1Z_2 + Z_2Z_3 + Z_3Z_1}{Z_2}$$

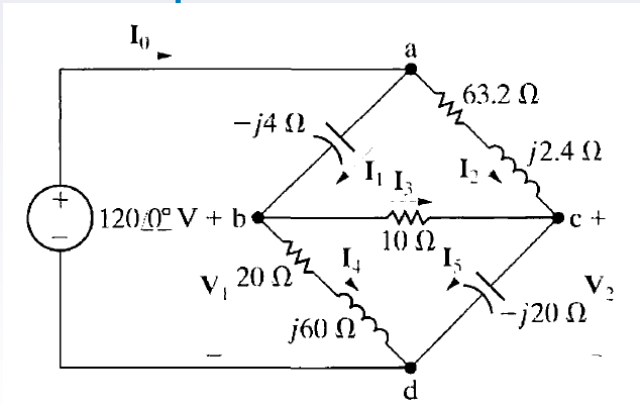
$$Z_c = \frac{Z_1Z_2 + Z_2Z_3 + Z_3Z_1}{Z_3}$$



Series, Parallel and Delta-to-Wye Simplifications

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Example-3: Use a Δ-to-Y impedance transformation to find $\boldsymbol{I_1, I_2, I_3, I_4, I_5, V_1}$ and $\boldsymbol{V_2}$.



Solution: First note that the circuit is not amenable to series or parallel simplification as it now stands. A Δ-to-Y impedance transformation allows us to solve for all the branch currents without resortng to either the node-voltage or the mesh-current method.

If we replace either the upper delta (abc) or the lower delta (bcd) with its Y equivalent, we can further simplify the resulting circuit by series-parallel combinations. In deciding which delta to replace, the sum of the impedances around each delta is worth checking because this quantity forms the denominator for the equivalent Y impedances. The sum around the lower delta is 30+j40, so we choose to eliminate it from the circuit. The Y impedances connecting to terminals b, c and d respectively are:

$$Z_1 = \frac{(20+j60)(10)}{30+j40} = 12 + j4\ \Omega \quad Z_2 = \frac{10(-j20)}{30+j40} = -3.2 - j2.4\ \Omega$$

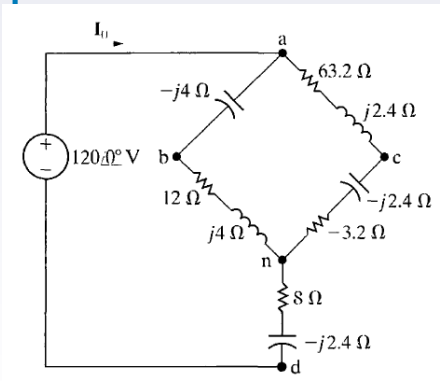
$$Z_3 = \frac{(20+j60)(-j20)}{30+j40} = 8 + j24\ \Omega$$

Series, Parallel and Delta-to-Wye Simplifications

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Example-3: Use a Δ-to-Y impedance transformation

to find $\boldsymbol{I_1}, \boldsymbol{I_2}, \boldsymbol{I_3}, \boldsymbol{I_4}, \boldsymbol{I_5}, \boldsymbol{V_1}$ and $\boldsymbol{V_2}$.



Continuation of the solution: Inserting the Y-equivalent impedances into the circuit, we get the circuit shown on the right, which we can now simplify by series-parallel reductions. The impedance of the abn branch and acn branch are:

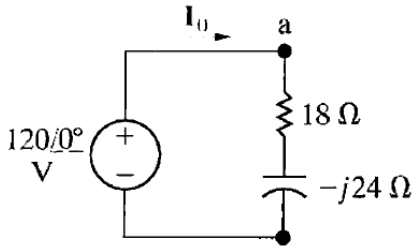
$$Z_{abn} = 12 + j4 - j4 = 12\ \Omega \quad Z_{acn} = 63.2 + j2.4 - j2.4 - 3.2 = 60\ \Omega$$

The abn branch is in parallel with the acn branch. We may replace these two branches with a single branch having their equivalent impedance of

$$Z_{an} = \frac{(60)(12)}{72} = 10\ \Omega$$

Combining this 10 Ω resistor with the impedance between n and d:

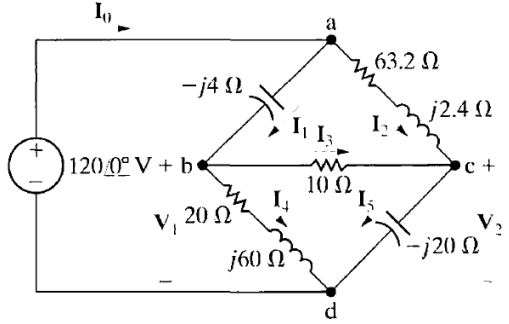
$$\boldsymbol{I_0} = \frac{120\angle 0^\circ}{18 - j24} = 4\angle 53.13^\circ = 2.4 + \text{j}3.2\ \text{A}$$



Series, Parallel and Delta-to-Wye Simplifications

Example-3: Use a Δ-to-Y impedance transformation

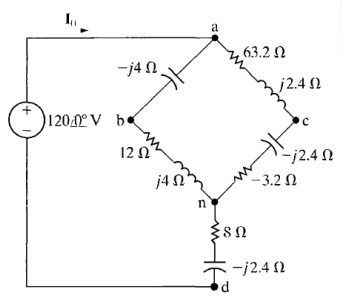
to find $\boldsymbol{I_1}, \boldsymbol{I_2}, \boldsymbol{I_3}, \boldsymbol{I_4}, \boldsymbol{I_5}, \boldsymbol{V_1}$ and $\boldsymbol{V_2}$.



Continuation of the solution: Working back through the equivalent circuits to find the branch currents in the original circuit:

$$\boldsymbol{V_{nd}} = (8 - j24)\boldsymbol{I_0} = 96 - j32\ V$$

$$\boldsymbol{V} = \boldsymbol{V_{an}} + \boldsymbol{V_{nd}} \qquad \boldsymbol{V_{an}} = 120 - 96 + j32 = 24 + j32\ V$$



We can now compute the branch currents $\boldsymbol{I_{abn}}$ and $\boldsymbol{I_{acn}}$:

$$\boldsymbol{I_{abn}} = \frac{24 + j32}{12} = 2 + j\frac{8}{3}\ A = \boldsymbol{I_1} \qquad \boldsymbol{I_{acn}} = \frac{24 + j32}{60} = \frac{4}{10} + j\frac{8}{15}\ A = \boldsymbol{I_2}$$

To find the branch currents $\boldsymbol{I_3}, \boldsymbol{I_4}$ and $\boldsymbol{I_5}$, we must first calculate $\boldsymbol{V_1}$ and $\boldsymbol{V_2}$.

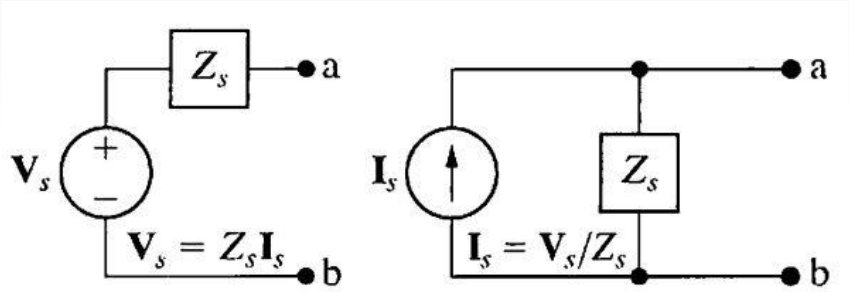
$$\boldsymbol{V_1} = 120\angle 0^\circ - (-\mathrm{j}4)\boldsymbol{I_1} = \frac{328}{3} + j8\ V \qquad \boldsymbol{V_2} = 120\angle 0^\circ - (63.2 + \mathrm{j}2.4)\boldsymbol{I_2} = 96 - j\frac{104}{3} V$$

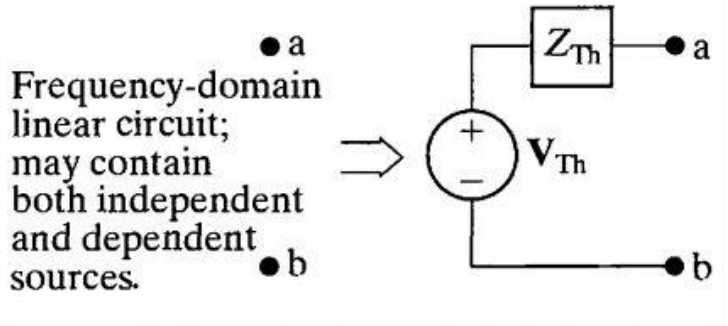
We can now calculate the branch currents $\boldsymbol{I_3}, \boldsymbol{I_4}$ and $\boldsymbol{I_5}$:

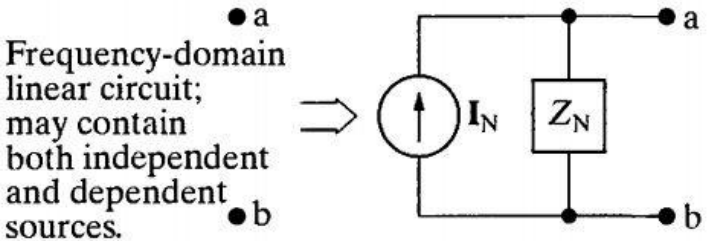
$$\boldsymbol{I_3} = \frac{\boldsymbol{V_1} - \boldsymbol{V_2}}{10} = \frac{4}{3} + j\frac{12.8}{3}\ A \qquad \boldsymbol{I_4} = \frac{\boldsymbol{V_1}}{20 + j60} = \frac{2}{3} + j1.6\ A \qquad \boldsymbol{I_3} = \frac{\boldsymbol{V_2}}{-\boldsymbol{j}20} = \frac{26}{15} + j4.8\ A$$

Source Transformations and Thévenin-Norton Equivalent Circuits

- The source transformations introduced for resistive circuits can be applied to frequency-domain circuits.



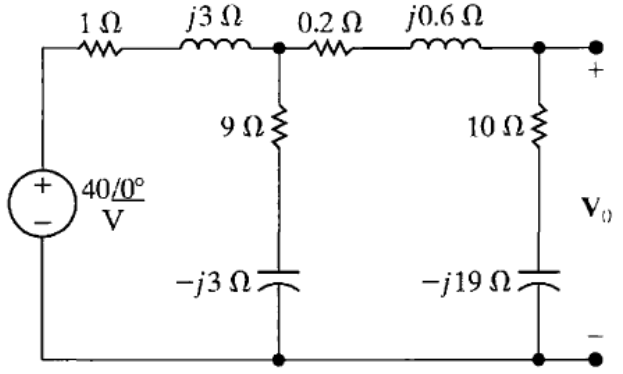




Source Transformations and Thévenin-Norton Equivalent Circuits

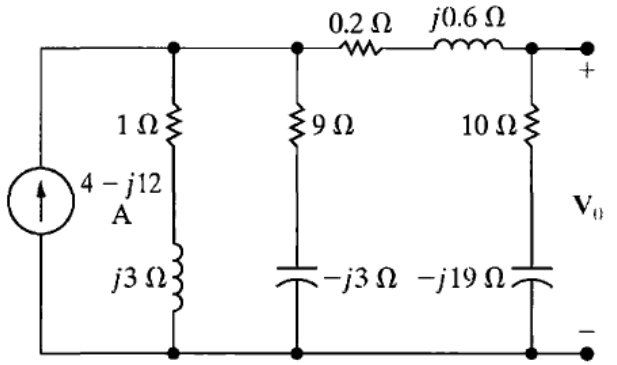
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Example-3: Use the concept of source transformation to find the phasor voltage $\boldsymbol{V_0}$ in the given circuit.



Solution: We can replace the series combination of the voltage source $(40\angle 0\ °)$ and the impedance $1 + j3\ \Omega$ with the parallel combination of a current source and the same impedance:

$$\boldsymbol{I} = \frac{40}{1+j3} = \frac{40}{10}(1 - j3) = 4 - j12\ A$$



Next we combine the two parallel branches into a single impedance

$$Z = \frac{(1+j3)(9-j3)}{10} = 1.8 + j2.4\ \Omega$$

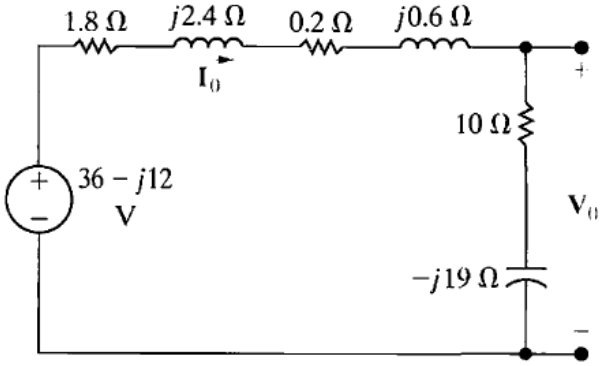
Another source transformation converts this parallel combination into a series combination.

$$\boldsymbol{V} = (4 - j12)(1.8 - j2.4) = 36 - j12\ V$$

Source Transformations and Thévenin-Norton Equivalent Circuits

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Example-3: Use the concept of source transformation to find the phasor voltage $\boldsymbol{V_0}$ in the given circuit.



Solution: We redraw the circuit. It is now a simple series circuit. We can calculate the current $\boldsymbol{I_0}$ by dividing the voltage of the source by the total series impedance:

$$\boldsymbol{I_0} = \frac{36 - j12}{12 - j16} = \frac{12(3 - j1)}{4(3 - j4)} = \frac{39 + j27}{25} = 1.56 + j1.08\ A$$

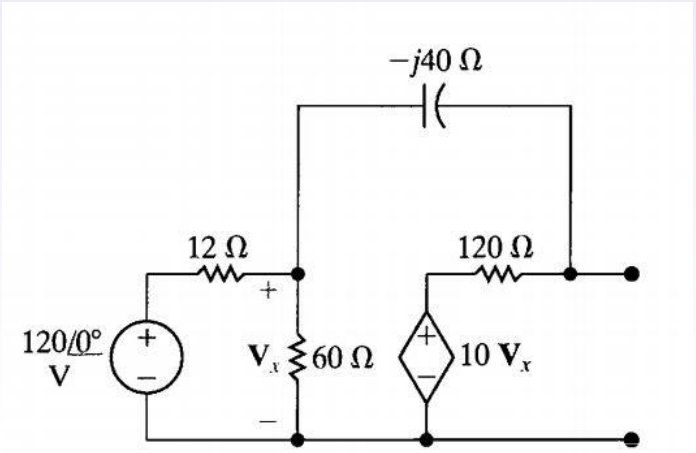
We can now obtain the value of $\boldsymbol{V_0}$ by multiplying $\boldsymbol{I_0}$ by the impedance $10 - j19$:

$$\boldsymbol{V_0} = (1.56 + j1.08)(10 - j19) = 36.12 - j18.84\ V$$

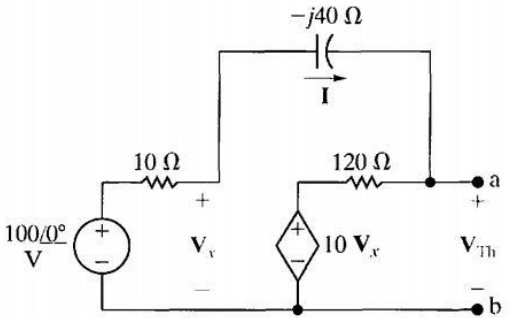
Source Transformations and Thévenin-Norton Equivalent Circuits

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Example-4: Find the Thévenin equivalent circuit with respect to terminals a, b in the given circuit.



Solution: We first determine the Thévenin equivalent voltage. It is the open-circuit voltage appearing at terminals a, b. We choose the reference for the Thévenin voltage as positive at terminal a. We can make two source transformation relative to the 120 V, 12 Ω and 60 Ω circuit elements to simplify this portion of the circuit. At the same time, these transformations must preserve the identity of the controlling voltage V_x, because of the dependent voltage. We first replace the series combination of the 120 V source and 12 Ω resistor with a 10 A current source in parallel with 12 Ω. Next we replace the parallel combination of the 12 and 60 Ω resistors with a single 10 Ω resistor. Finally, we replace the 10 A source in parallel with 10 Ω with a 100 V source in series with 10 Ω.

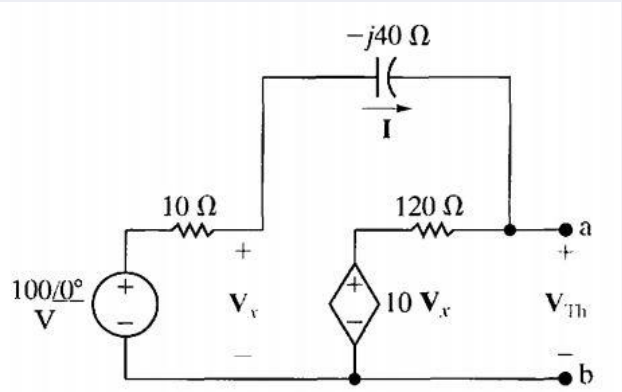


Source Transformations and Thévenin-Norton Equivalent Circuits

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Example-4: Once we know the current $\boldsymbol{I}$ we can compute the Thévenin voltage. We find $\boldsymbol{I}$ by summing the voltage around the closed path.

$$100 = 10\boldsymbol{I} - j40\boldsymbol{I} + 120\boldsymbol{I} + 10\boldsymbol{V_x} = (130 - j40)\boldsymbol{I} + 10\boldsymbol{V_x}$$



We can also write

$$\boldsymbol{V_x} = 100 - 10\boldsymbol{I}$$

Then

$$\boldsymbol{I} = \frac{-900}{30 - j40} = 18\angle -126.87°\ A$$

We can now calculate $\boldsymbol{V_x}$: $\boldsymbol{V_x} = 100 - 10\boldsymbol{V_x} = 208 + j144\ V$

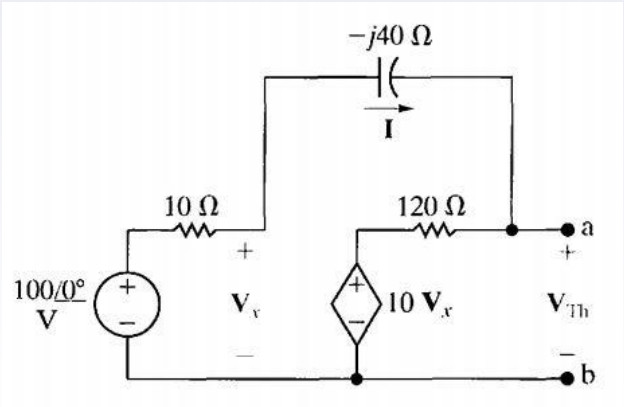
Using the relation

$$\boldsymbol{V_{Th}} = 10\boldsymbol{V_x} + 120\boldsymbol{I} = 2080 + j1440 + 120(18)\angle -126.87°$$
$$= 784 - j288 = 835.22\angle -20.17°$$

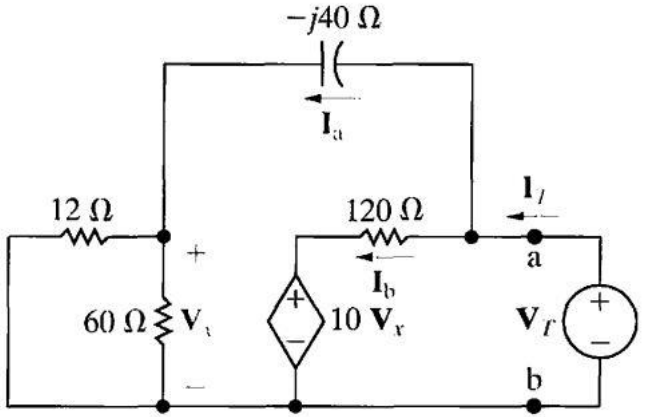
Source Transformations and Thévenin-Norton Equivalent Circuits

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Example-4: To find the Thévenin impedance, we may use test-source method. We deactivate all independent sources from the circuit and then apply either a test voltage source or a test current source to the terminals of interest.
The ratio of the voltage to the current at the source is the Thévenin impedance.



The branch current I_a and I_b have been added to the circuit to smiplify the calculation of I_T.



$$I_a = \frac{V_T}{10 - j40} \qquad V_x = 10 I_a$$

$$I_b = \frac{V_T - 10V_x}{120} = \frac{-V_T(9 + j4)}{120(1 - j4)}$$

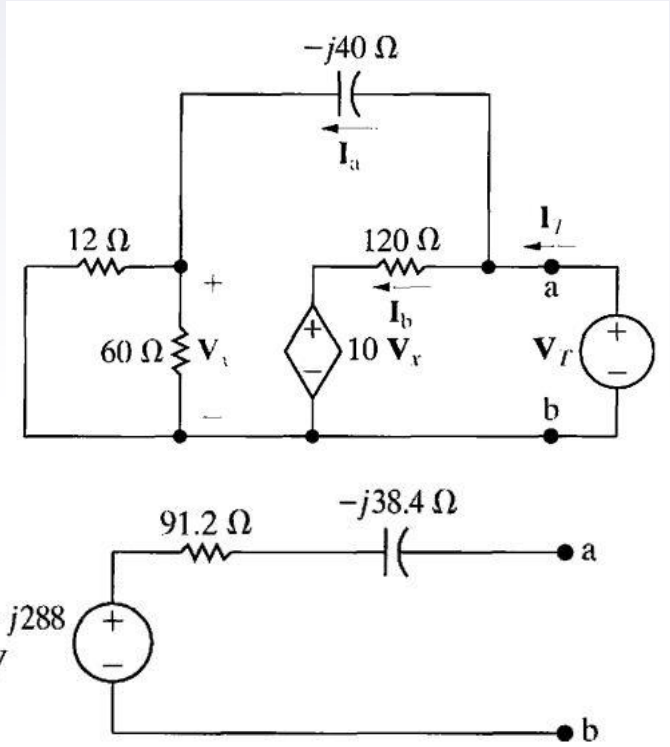
$$I_T = I_a + I_b = \frac{V_T}{10 - j40}\left(1 - \frac{(9 + j4)}{12}\right) = \frac{-V_T(3 - j4)}{12(10 - j40)}$$

Source Transformations and Thévenin-Norton Equivalent Circuits

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Example-4:

$$Z_{th} = \frac{\boldsymbol{V_T}}{\boldsymbol{I_T}} = 91.2 - j38.4\ \Omega$$



The Node-Voltage Method

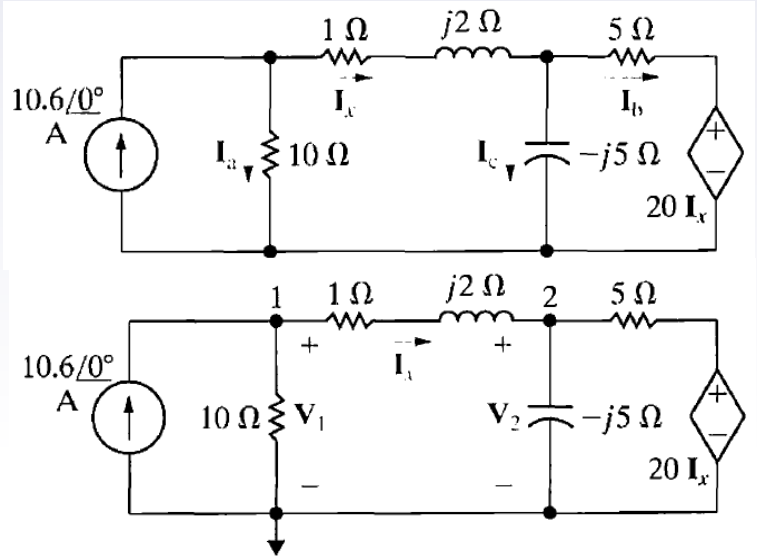
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- The same concepts of the node-voltage method previously introduced for resistive circuits apply when we use the node-voltage method to analyze frequency-domain circuits
- It is better to show it with an example.

The Node-Voltage Method

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Example-5: Use the node-voltage method to find the branch currents $\boldsymbol{I_a}$, $\boldsymbol{I_b}$ and $\boldsymbol{I_c}$ in the given circuit.



Solution: There are three essential nodes. Two equations can be written.Taking the the bottom node as the reference node.

For node 1, we can write

$$-10.6 + \frac{\boldsymbol{V_1}}{10} + \frac{\boldsymbol{V_1} - \boldsymbol{V_2}}{1 + j2} = 0$$

By multiplying by 1+j2 and collecting the coefficients of $\boldsymbol{V_1}$ and $\boldsymbol{V_2}$

$$\boldsymbol{V_1}(1.1 + j0.2) - \boldsymbol{V_2} = 10.6 + j21.2$$

For node 2, we can write

$$\frac{\boldsymbol{V_2} - \boldsymbol{V_1}}{1 + j2} + \frac{\boldsymbol{V_2}}{-j5} + \frac{\boldsymbol{V_2} - 20\boldsymbol{I_X}}{5} = 0$$

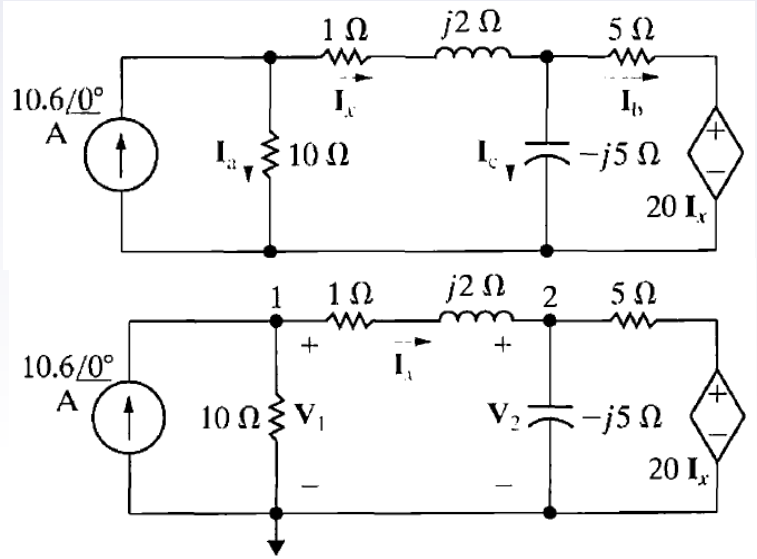
The controlling current $\boldsymbol{I_X}$ is: $\boldsymbol{I_X} = \frac{\boldsymbol{V_1} - \boldsymbol{V_2}}{1 + j2}$

The Node-Voltage Method

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Example-5: Substituting expression for $\boldsymbol{I_X}$ into the node 2 equation, multiplying by 1+j2, and collecting coefficients of $\boldsymbol{V_1}$ and $\boldsymbol{V_2}$ produces the equation

$$5\boldsymbol{V_1} + (4.8 + j0.6)\boldsymbol{V_2} = 0$$



The solutions for $\boldsymbol{V_1}$ and $\boldsymbol{V_2}$ are

$$\boldsymbol{V_1} = 68.40 - j16.80\ V$$

$$\boldsymbol{V_2} = 68 - j26\ V$$

The branch currents are

$$\boldsymbol{I_a} = \frac{\boldsymbol{V_1}}{10} = 6.84 - j1.68\ A \qquad \boldsymbol{I_X} = \frac{\boldsymbol{V_1} - \boldsymbol{V_2}}{1+j2} = 3.76 + j1.68\ A$$

$$\boldsymbol{I_b} = \frac{\boldsymbol{V_2} - 20\boldsymbol{I_X}}{5} = -1.44 - j11.92\ A \qquad \boldsymbol{I_c} = \frac{\boldsymbol{V_2}}{-j5} = 5.2 - j13.6\ A$$

The Mesh-Current Method

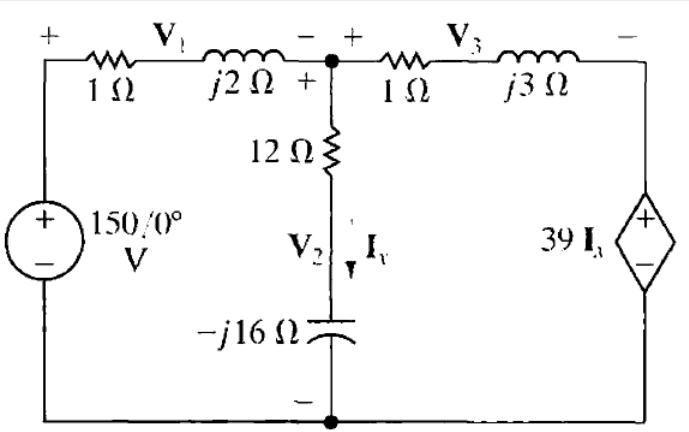
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- The same concepts of the mesh-current method previously introduced for resistive circuits apply when we use the mesh-mesh current method to analyze frequency-domain circuits
- It is better to show it with an example.

The Mesh-Current Method

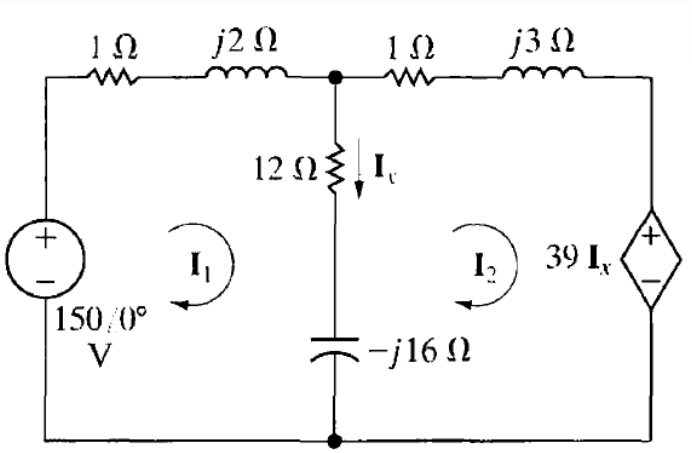
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Example-5: Use the mesh-current method to find the voltages $\boldsymbol{V_1}$, $\boldsymbol{V_2}$ and $\boldsymbol{V_3}$ in the given circuit.



Solution: The circuit has two meshes and a dependent voltage source, so we must write two mesh-current equations and a constraint equation.

The reference direction for mesh currents $\boldsymbol{I_1}$ and $\boldsymbol{I_2}$ is clockwise. Once we know $\boldsymbol{I_1}$ and $\boldsymbol{I_2}$, we can easily find the unknown voltages.



For mesh 1

$$150 = (1 + j2)\boldsymbol{I_1} + (12 - \mathrm{j}16)(\boldsymbol{I_1} - \boldsymbol{I_2})$$

$$150 = (13 + j14)\boldsymbol{I_1} - (12 - \mathrm{j}16)\boldsymbol{I_2}$$

For mesh 2

$$0 = (12 - j16)(\boldsymbol{I_2} - \boldsymbol{I_1}) + (1 + \mathrm{j}3)\boldsymbol{I_2} + 39\boldsymbol{I_X}$$

$$\boldsymbol{I_X} = \boldsymbol{I_1} - \boldsymbol{I_2}$$

The Mesh-Current Method

Example-5:

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Substituting the constaint into the mesh 2 equation and simplifying the resulting expression

$$0 = (27 + j16)\boldsymbol{I_1} + (26 + j13)\boldsymbol{I_2}$$

Solving for $\boldsymbol{I_1}$ and $\boldsymbol{I_2}$

$$\boldsymbol{I_1} = -26 - j52\ A$$

$$\boldsymbol{I_2} = -24 - j58\ A$$

$$\boldsymbol{I_X} = -2 + j6\ A$$

The three voltages are

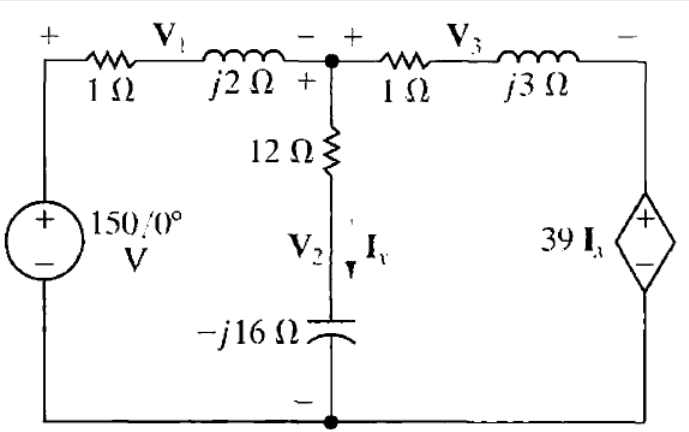
$$\boldsymbol{V_1} = (1 + j2)\boldsymbol{I_1} = 78 - j104\ V$$

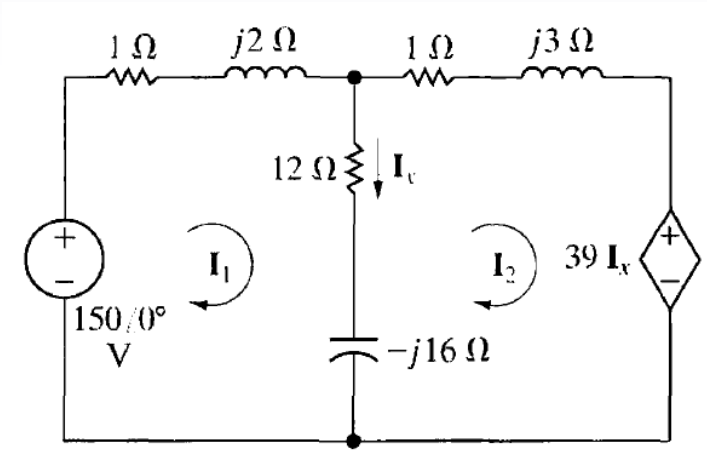
$$\boldsymbol{V_2} = (12 - j16)\boldsymbol{I_X} = 72 + j104\ V$$

$$\boldsymbol{V_3} = (1 + j3)\boldsymbol{I_2} = 150 - j130\ V$$

Also

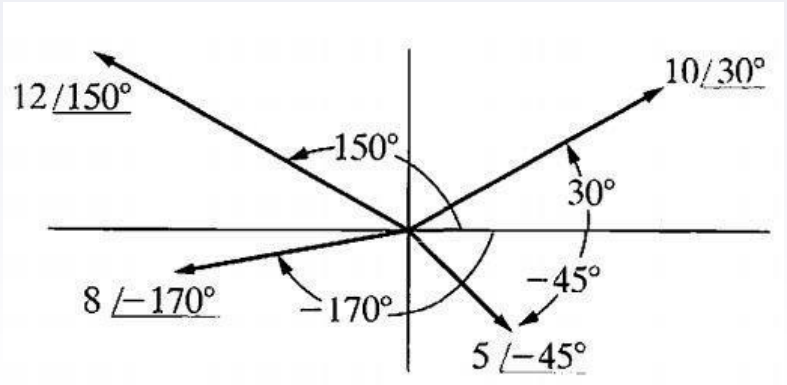
$$39\boldsymbol{I_X} = -78 + j234\ V$$

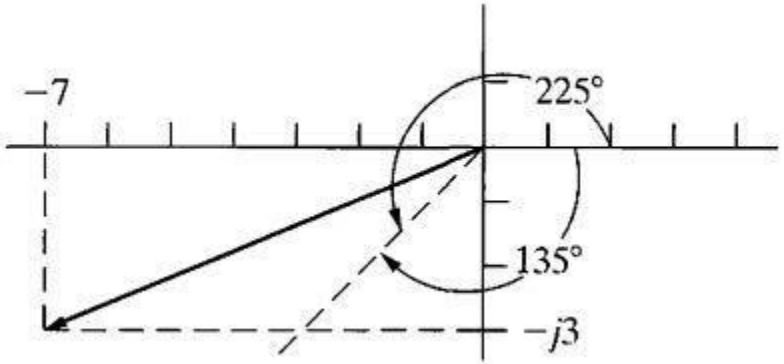




Phasor Diagrams

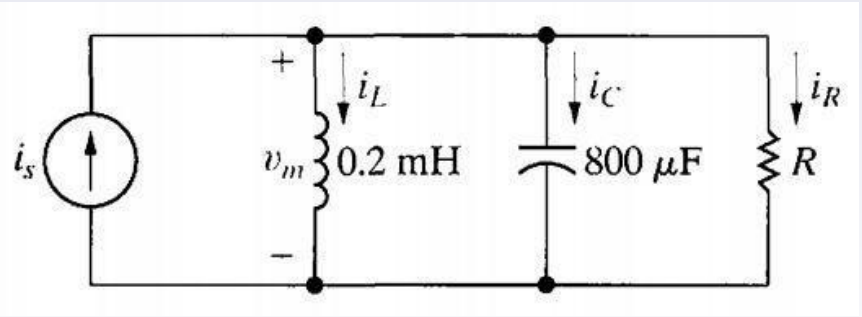
- When using the phasor method to analyze the steady-state sinusoidal operation of a circuit, a diagram of the phasor currents and voltages may give further insight into the behavior of the circuit.
- A phasor diagram shows the magnitude and the phase angle of each phasor quantitiy in the complex-number plane.
- Phase angles are measured counterclockwise from the positive real axis, and magnitudes are measured from the origin of the axes.
- Constructing phasor diagrams of circuit quantities, generally involves both currents and voltages.(two different magnitude scales are used in such cases).





Phasor Diagrams

Example-1: For the given circuit, use a phasor diagram to find the value R that will cause the current through that resistor, i_R, to lag the source current, i_s, by 45°, when $\omega = 5\ krad/s$



Solution: By KCL, the sum of the currents $\boldsymbol{I}_R$, $\boldsymbol{I}_L$ and $\boldsymbol{I}_C$ must equal the source current $\boldsymbol{I}_s$. If we assume that the phase angle of the voltage $\boldsymbol{V}_m$ is zero, we can draw the current phasors for each of the components. The current phasor for the inductor:

$$\boldsymbol{I}_L = \frac{V_m \angle 0°}{j(5000)(0.2 \times 10^{-3})} = V_m \angle -90°$$

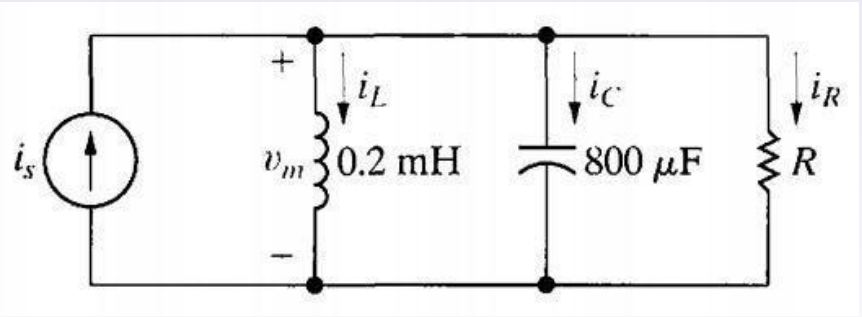
The current phasor for the capacitor:

$$\boldsymbol{I}_C = \frac{V_m \angle 0°}{-j/[(5000)(800 \times 10^{-6})]} = 4V_m \angle 90°$$

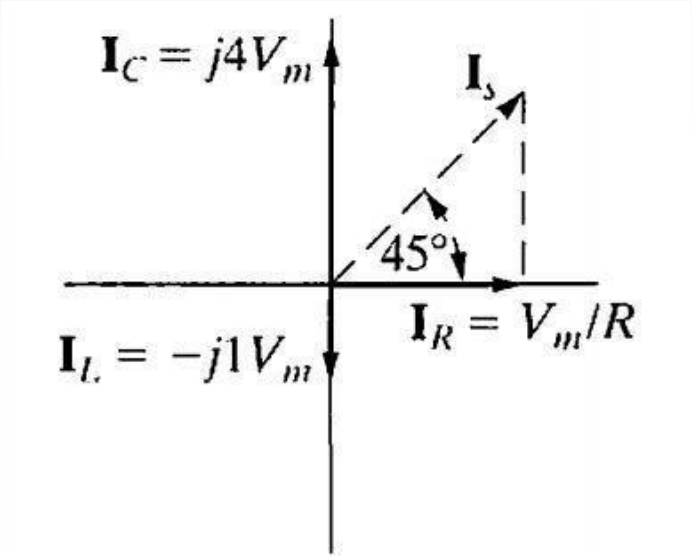
And the current phasor for the resistor:

$$\boldsymbol{I}_R = \frac{V_m \angle 0°}{R} = \frac{V_m}{R} \angle 0°$$

Phasor Diagrams

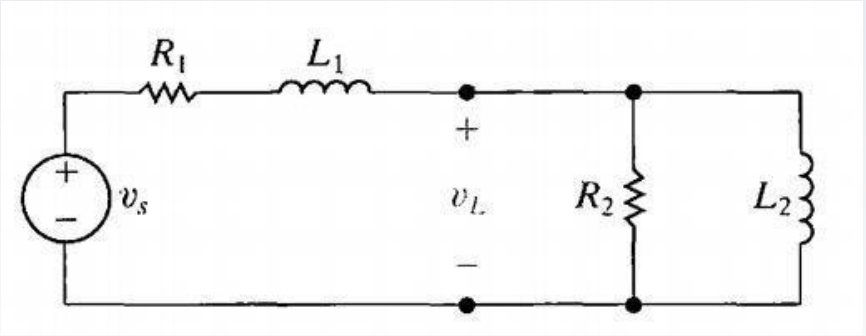


Example-1: For the given circuit, use a phasor diagram to find the value R that will cause the current through that resistor, i_R, to lag the source current, i_s, by 45°, when $\omega = 5\ krad/s$



Solution: The phasors are shown in the diagram. It also shows thesource current phasor, sketched as a dotted line, which must be thesum of the current phasors of the three circuit components andmust be at an angle that is 45° more positive than the current phasor of the resistor. As you can see, summing the phasors makes an isosceles triangle, so the length of the current phasor for the resistor must equal $3V_m$. Therefore the value of the resistor is $\frac{1}{3}\ \Omega$

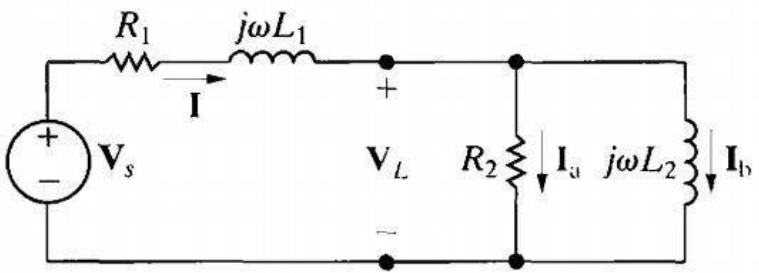
Phasor Diagrams



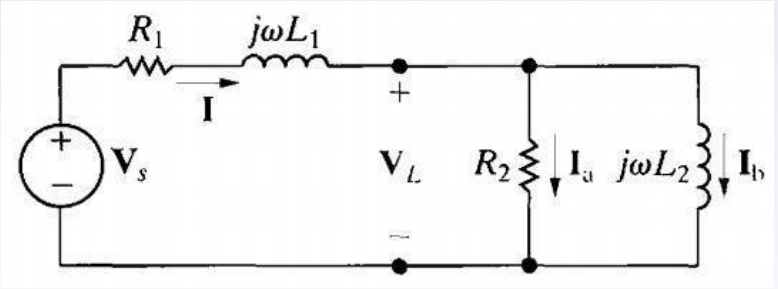
Example-2: The given circuit has a load consisting of the parallel combination of the resistor and inductor. Use phasor diagrams to explore the effect of adding a capacitor accross the terminals of the load on the amplitude of $\boldsymbol{V}_s$, if we adjust $\boldsymbol{V}_s$ so that the amplitude of $\boldsymbol{V}_L$ remains constant. Utility companies use this techniques to control the voltage drop on their lines.

Solution: We begin by assuming zero capacitance across the load. After constructing the phasor diagram for the zero-capacitance case, we can add the capacitor and study its effect on the amplitude of $\boldsymbol{V}_s$, holding the amplitude of $\boldsymbol{V}_L$ constant.

The frequency-domain equivalent of the circuit (also adding the phasor branch current $\boldsymbol{I}$, $\boldsymbol{I}_a$ and $\boldsymbol{I}_b$:



Phasor Diagrams



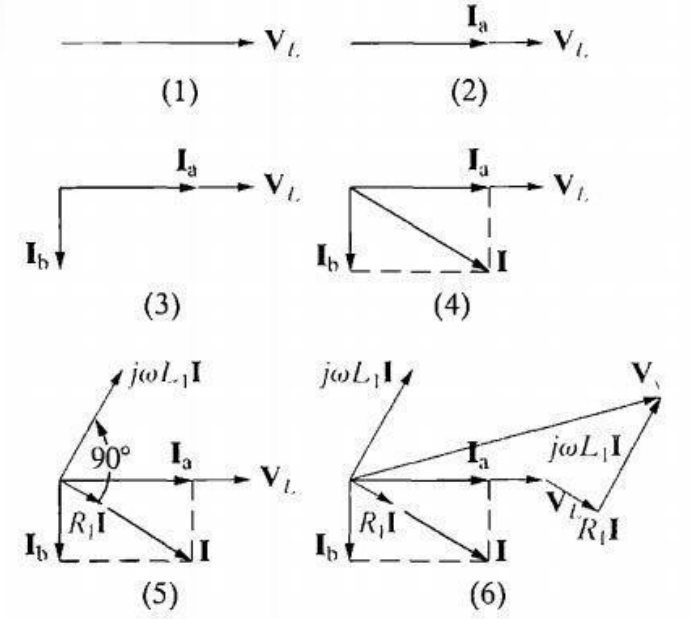
Solution: We can draw the phasor diagram step by step. Keep in mind that we are not interested in specific phasor values and positions in this example, but rather in the general effect of adding a capacitor across the terminals of the load. Thus, we want to develop the rrelative positions of the phasors before and after the capacitor has been added.

1- Because we are holding the amplitude of the load voltage constant, we choose $\boldsymbol{V}_L$ as our reference.

2- We know that $\boldsymbol{I}_a$ is in phase with $\boldsymbol{V}_L$ and that its magnitude is $|\boldsymbol{V}_L|/R_2$ (on the phasor diagram, the magnitude scale for the current phasors is independent of the magnitude scale for the voltage phasors).

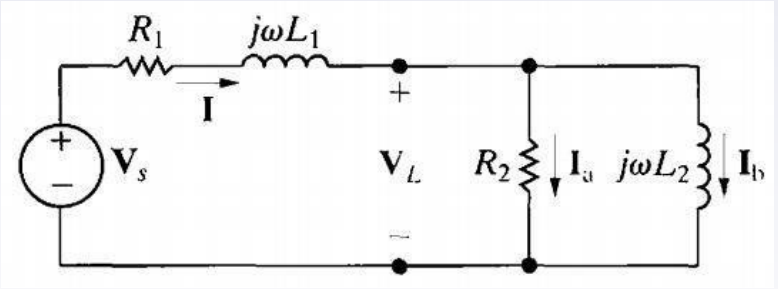
3- We know that $\boldsymbol{I}_b$ lags behind $\boldsymbol{V}_L$ by 90° and that its magnitude is $|\boldsymbol{V}_L|/\omega L_2$.

4- The line current $\boldsymbol{I}$ equal to the sum of $\boldsymbol{I}_a$ and $\boldsymbol{I}_b$.



Phasor Diagrams

Solution:.

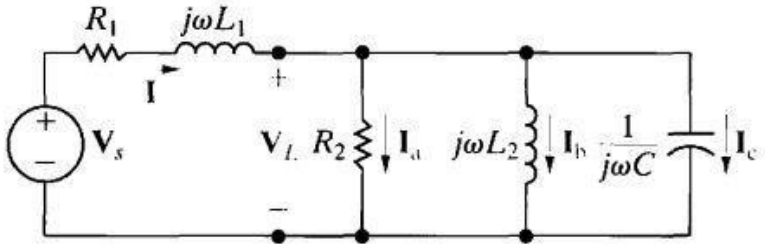


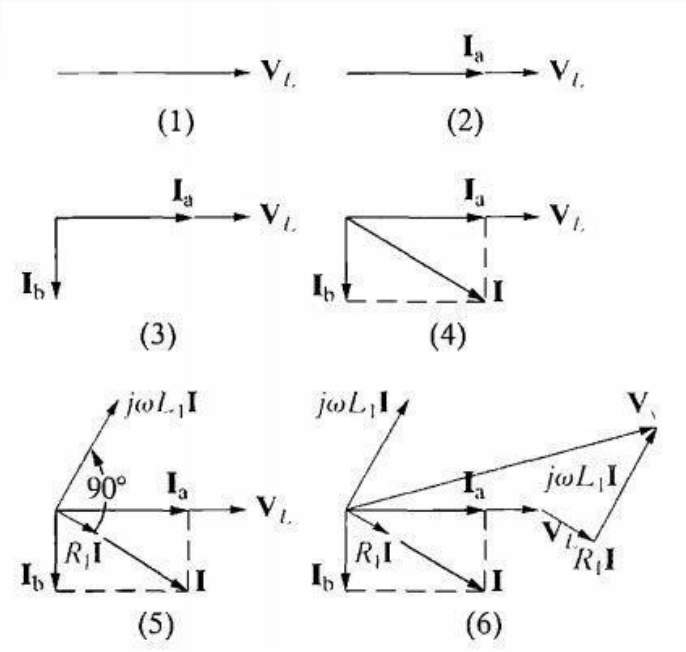
5The voltage drop across R_1 is in phase with the line current, and the voltage drop across $j\omega L_1$ leads the line current by 90°.

6The source voltage is the sum of the load voltage and the drop along the lines; that is, $\boldsymbol{V}_s = \boldsymbol{V}_L + (R_1 + j\omega L_1)\boldsymbol{I}$

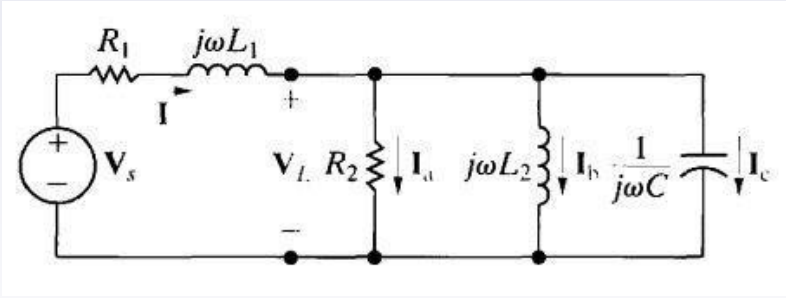
The completed phasor diagram clearly shows the amplitude and phase angle relationships among all the current and voltages.

Now we add the capacitor branch as shown below:



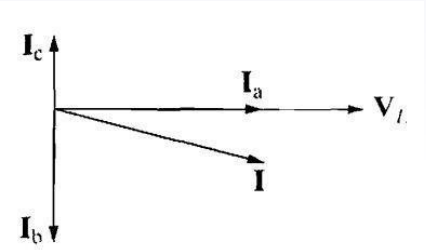


Phasor Diagrams



Solution:.

We are holding $\boldsymbol{V}_L$, so we construct the phasor diahram for the circuit following the same steps, except that, in step 4, we add the capacitor current $\boldsymbol{I}_c$ to the diagram. In so doing, $\boldsymbol{I}_c$ leads $\boldsymbol{V}_L$ by 90°, with its magnitude being $|\boldsymbol{V}_L \omega C|$.

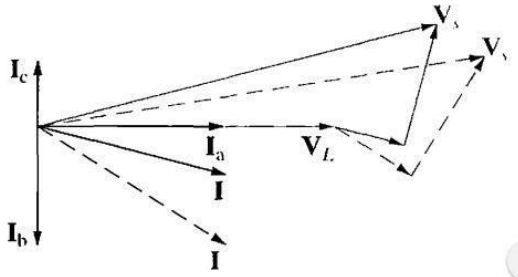


The figure at the right shows the effect of $\boldsymbol{I}_c$ on the line current:

Both the magnitude and phase angle of line current $\boldsymbol{I}$ change with changes in the magnitude of $\boldsymbol{I}_c$.

As $\boldsymbol{I}$ changes, so do the magnitude and phase angle of the voltage drop along the line. As the drop along the line changes, the magintude and phase angle of $\boldsymbol{V}_s$ change. The phasor diagram depicts these observations. The dotted phasors represent the pertinent current and voltages before the addition of the capacitor.

Comparing the dotted phasors of $\boldsymbol{I}$, $R_1\boldsymbol{I}$, $j\omega L_1\boldsymbol{I}$, and $\boldsymbol{V}_s$ with their solid counterparts, adding C to the circuit reduces the amplitude of the source voltage and still maintains the amplitude of the load voltage. As the load increases, we can add capacitors to the system to maintain $\boldsymbol{V}_L$ without increasing the amplitude of the source voltage.



EE2001 Electrical Circuits- I:

Sinusoidal Steady-State Power Calculations

08 January 2024

Outline



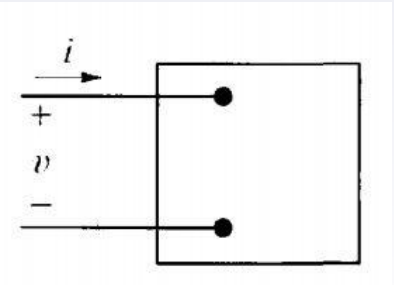
1. Instantaneous Power
2. Average and Reactive Power
3. The rms Value and Power Calculations
4. Complex Power
5. Power Calculations
6. Maximum Power Transfer

Instantaneous Power

1

- Considering v and i are steady-state sinusoidal signals, using the passive sign convention, the power at any instant of time is:

$$p = vi$$



- This is instantaneous power.
- The expressions for v and i

$$v = V_m \cos(\omega t + \theta_v)$$
$$i = I_m \cos(\omega t + \theta_i)$$

where θ_v is the voltage phase angle and θ_i is the current phase angle.

- We are operating in the sinusoidal-steady state, so we may choose any convenient reference for zero time. A zero time corresponding to the instant the current is passing through a positive maximum is usually chosen. This reference system requires shift of both the voltage and the current by θ_i. Thus equations become:

$$v = V_m \cos(\omega t + \theta_v - \theta_i)$$
$$i = I_m \cos \omega t$$

Instantaneous Power

1

- Substituting the voltage and current equations into the expression for the instantaneous power:

$$p = V_m I_m \cos(\omega t + \theta_v - \theta_i) cos\omega t$$

- We can further edit this equation considering the trigonometric identity:

$$cos\alpha\ cos\beta = \frac{1}{2}\cos(\alpha - \beta) + \frac{1}{2}\cos(\alpha + \beta)$$

- Letting $\alpha = \omega t + \theta_v - \theta_i$ and $\beta = \omega t$ gives

$$p = \frac{V_m I_m}{2}\cos(\theta_v - \theta_i) + \frac{V_m I_m}{2}\cos(2\omega t + \theta_v - \theta_i)$$

- Using the trigonometric identity

$$\cos(\alpha + \beta) = cos\alpha\ cos\beta - sin\alpha\ sin\beta$$

- Applying that to the second term on the right-hand side of power equation:

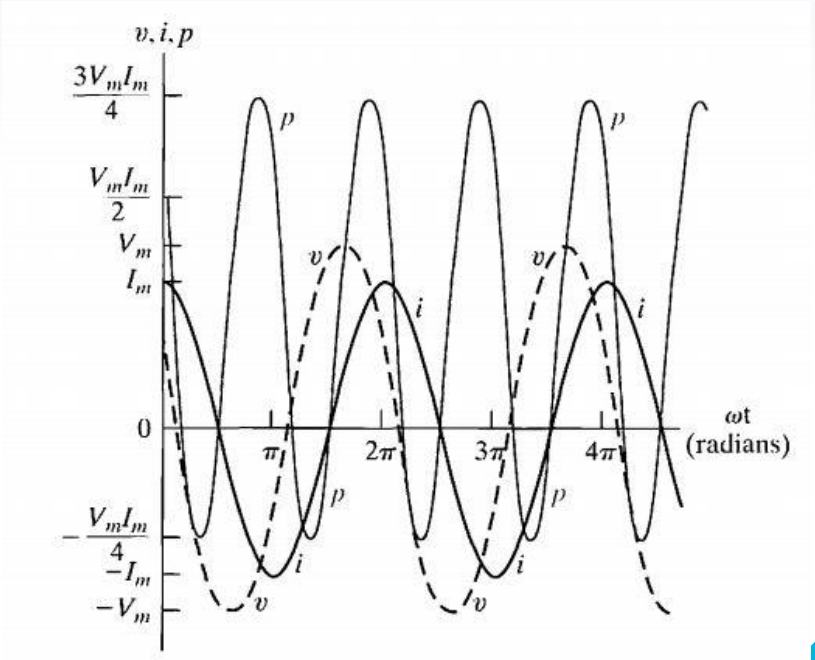
$$p = \frac{V_m I_m}{2}\cos(\theta_v - \theta_i) + \frac{V_m I_m}{2}\cos(\theta_v - \theta_i)\, cos2\omega t - \frac{V_m I_m}{2}\sin(\theta_v - \theta_i)\, sin2\omega t$$

Instantaneous Power

1

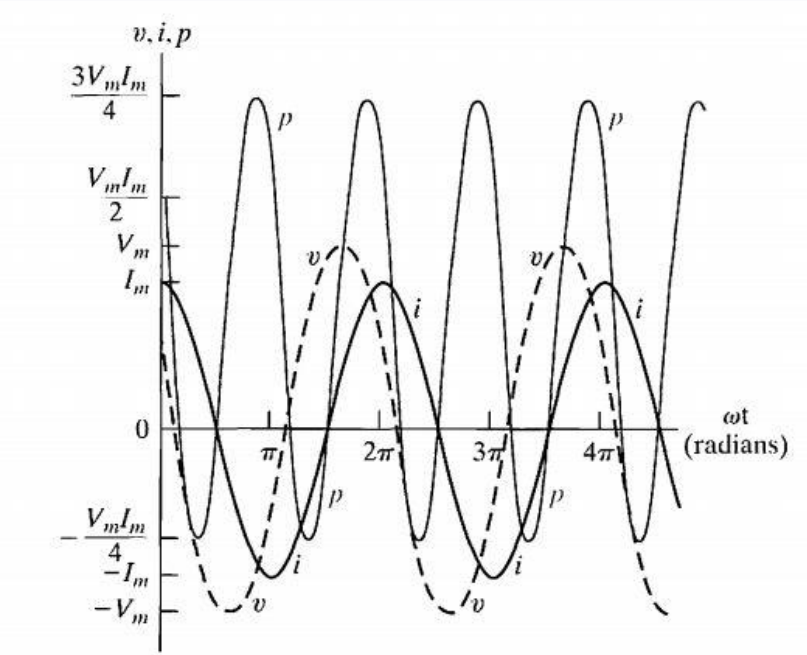
$$p = \frac{V_m I_m}{2}\cos(\theta_v - \theta_i) + \frac{V_m I_m}{2}\cos(\theta_v - \theta_i)\,cos2\omega t - \frac{V_m I_m}{2}\sin(\theta_v - \theta_i)\,sin2\omega t$$

- Depicting a representative relationship among v, i and p, for an example case with $\theta_v = 60°$ and $\theta_i = 0°$:
- From the figure it can be observed that the frequency of the instataneous power is twice the frequency of the voltage or current.
- Therefore, the instantaneous power goes through two complete cycles for every cycle of either the voltage or the current.
- Also note that the instantaneous power may be negative for a portion of each cycle, even if the network between the terminals is passive.



Instantaneous Power

- In a completely passive network, negative power implies that energy stored in the inductors or capacitors is now being extracted.
- Variation of instataneous power with time in the sinusoidal steady-state operation of a circuit explains why some motor-driven applications experience vibration and require resilient motor mountings.



Average Power

2

- We can write the instantaneous power equation as below:

$$p = P + Pcos2\omega t - Qsin2\omega t$$

where

P is called the average (real, active) power:

$$P = \frac{V_m I_m}{2}\cos(\theta_v - \theta_i)$$

and Q is called the reactive power:

$$Q = \frac{V_m I_m}{2}\sin(\theta_v - \theta_i)$$

- Average power or real power, describes the power in a circuit that is transformed from electric to nonelectric energy.

Average Power

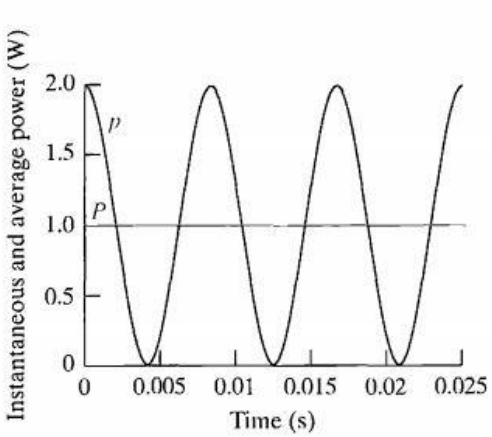
2

Power for Purely Resistive Circuits

- If the circuit between the terminals is purely resistive, the voltage and current are in phase, which means that $\theta_v = \theta_i$. Then the instantaneous power equation reduces to

$$p = P + Pcos2\omega t$$

and becomes instantaneous real power equation.



- An example graph of a representative purely resistive circuit with $\omega = 377\ rad/s$ and $P = 1\ W$
- In such as case, instantaneous real power can never be negative, power cannot be extracted from a purely resistive network. All the electric energy is dissipated in the form of thermal energy.

Average Power

2

Power for Purely Inductive Circuits

- If the circuit between the terminals is purely inductive, the voltage and current are out of phase by precisely 90°. The current lags the voltage by 90° (that is, $\theta_i = \theta_v - 90°$). Then the instantaneous power equation reduces to

$$p = -Qsin2\omega t$$

- In a purely inductive circuit, the average power (real power) is zero. There is no transformation of energy from electric to nonelectric form.
- In such a case, the power is continually exchanged between the circuit and the source driving the circuit, at frequency of 2ω.
- In other words, when p is positive, energy is stored in the magnetic fields associated with the inductive elements, and when p is negative, energy is being extracted from the magnetic fields.

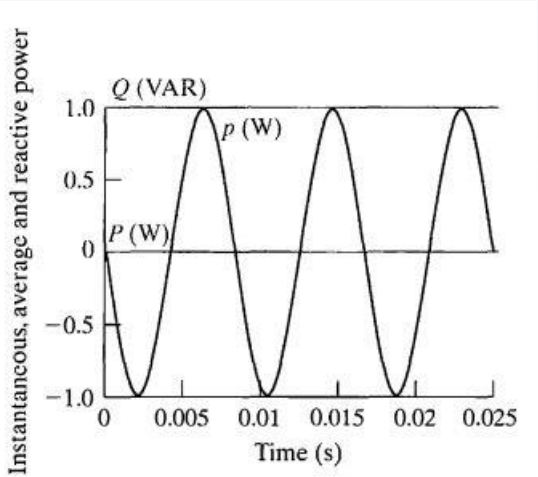
Average Power

2

Power for Purely Inductive Circuits

$$p = -Qsin2\omega t$$

- A measure of the power associated with purely inductive circuits is the reactive power Q.
- The name reactive power comes from the characterization of an inductor as a reactive elements; its impedance is purely reactive.
- Note that the average power P and reactive power Q carry the same dimension. To distinguish between them we use the units watt (W) for average power and VAr (volt-amp reactive) for reactive power.



- An example graph of a representative purely inductive circuit with $\omega = 377\ rad/s$ and $Q = 1\ VAr$

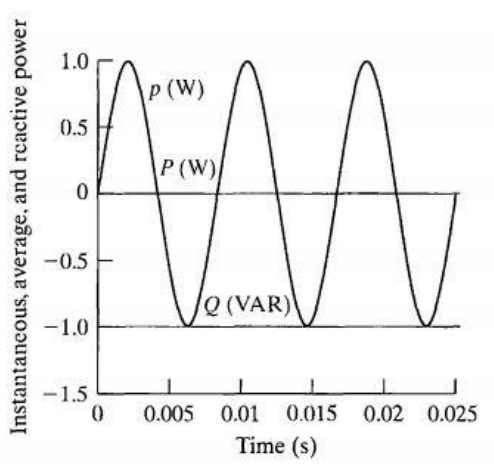
Average Power

Power for Purely Capacitive Circuits

2

- If the circuit between the terminals is purely capacitive, the voltage and current are out of phase by precisely 90°. The current leads the voltage by 90° (that is, $\theta_i = \theta_v + 90°$). Then the instantaneous power equation reduces to

$$p = -Qsin2\omega t$$



- Again, the average power (real power) is zero. There is no transformation of energy from electric to nonelectric form.
- In such a case, the power is continually exchanged between the circuit and the source driving the circuit and the electric field associated with the capacitive elements. An example graph of a representative purely capacitive circuit with $\omega = 377\ rad/s$ and $Q = -1\ VAr$
- The decision to use the current as the reference leads to Q being positive for inductors and negative for capacitors. Power engineers recognize this difference in the algebraic sign of Q by saying that inductors demand (or absorb) reactive power and capacitors supply (deliver) reactive power.

Average Power

The Power Factor

2

- The angle $\theta_v - \theta_i$ plays a role in the computation of both active and reactive power and is referred to as the power factor angle.
- The cosine of this angle is called the power factor, abbreviated pf, and the sine of this angle is called the reactive factor, abbreviated rf. Thus

$$\text{pf} = \cos(\theta_v - \theta_i)$$

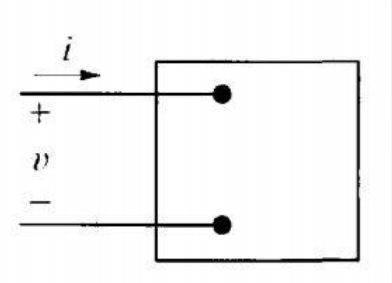
$$\text{rf} = \sin(\theta_v - \theta_i)$$

- Knowing the value of the power factor does not tell you the value of the power factor angle, because $\cos(\theta_v - \theta_i) = \cos(\theta_i - \theta_v)$. To completely describe this angle, we use the descriptive phrases **lagging power factor** and **leading power factor**.
- Lagging power factor implies that current lags voltage – hence an inductive load.
- Leading power factor implies that current leads voltage – hence a capacitive load.
- Both power factor and reactive factor are convenient quantities to use in describing electrical loads.

Average Power

2

Example: a) Calculate the average (real, active) power and the reactive power at the terminals of the given network if



$$v = 100\cos(\omega t + 15°)\,V$$
$$i = 4\sin(\omega t - 15°)\,A$$

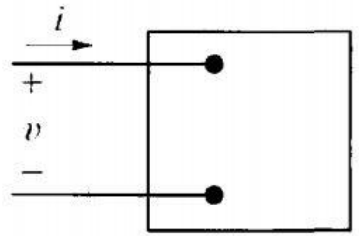
Solution a) Because i is expressed in terms of the sine functions, the first step is to rewrite i as a cosine function:

$$i = 4\,cos(\omega t - 105°)\,A$$

We now calculate P and Q directly from general equations:

$$P = \frac{1}{2}(100)(4)cos[15 - (-105)] = -100\,W$$
$$Q = \frac{1}{2}(100)(4)sin[15 - (-105)] = 173.21\,VAr$$

Average Power



2 Example: b) State whether the network inside the box absorbing or delivering average power.

c)State whether the network inside the box is absorbing or delivering magnetizing reactive power.

d) Calculate the power factor and reactive factor values.

Solution b) We used the passive sign convention with a positive sign in the power formula. Getting a negative result in this case (-100W) means that the network inside the box is delivering (supplying) average power to the terminals.

c) The passive sign conventions means that, because the Q value we found is positive, the network inside the box is absorbing (demanding) reactive power at its terminals.

d) $\text{pf} = \cos(\theta_v - \theta_i - 180) = \cos[15 - (-105 + 180)] = \cos(15 - 75) = \cos(-60) = 0.5$

$\text{rf} = \sin(-60) = -0.866$

The rms Value and Power Calculations

3

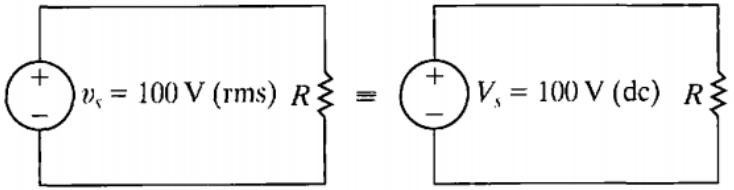
- For a resistive circuit:

$$P = \frac{V_{rms}^{\ 2}}{R}$$

$$P = I_{rms}^{\ 2} R$$

- The rms value is also referred to as the effective value of the sinusoidal voltage (or current).
- Given an equivalent resistive load R, and an equivalent time period T, the rms value of a sinusoidal source delivers the same energy to R as does a DC source of the same value.
- Average power in terms of effective values:

$$P = \frac{V_m I_m}{2} cos(\theta_v - \theta_i) = \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} cos(\theta_v - \theta_i) = V_{eff} I_{eff} cos(\theta_v - \theta_i)$$



The rms Value and Power Calculations

3

- In a similar manner:

$$Q = V_{eff} I_{eff} sin(\theta_v - \theta_i)$$

- Effective value of the sinusoidal signal in power calculations is so widely used that voltage and current ratings of circuits and equipment in power utilization are given in terms of rms values.

Example: A 120 V, 100 W lamps has a resistance of $120^2/100 = 144\ \Omega$ and draws an rms current of $120/144 = 0.833\ A$. The peak value of the lamp current is $0.833\sqrt{2} = 1.18\ A$.

- The phasor transform of a sinusoidal function may also be expressed in terms of the rms value. The magnitude of the rms phasor is equal to the rms value of the sinusoidal function. If a phasor is based on the rms value, we indicate this by either an explicit statement, a parenthetical ‘‘ rms’’ adjacent to the phasor quantitiy, or the subscript ‘‘ eff’’.

The rms Value and Power Calculations

3

Example: a) A sinusoidal voltage having a maximum amplitude of 625 V is applied to the terminals of a 50 Ω resistor. Find the average power delivered to the resistor.

Solution a: The rms value of the sinusoidal voltage is $625/\sqrt{2} = 441.94\ V$

The average power delivered to the 50 Ω resistor is

$$P = \frac{V_{rms}{}^2}{R} = \frac{(441.94)^2}{50} = 3906.25\ W$$

Example: b) Repeat (a) by first finding the current in the resistor.

Solution b: The maximum amplitude of the current in the resistor is $625/50 = 12.5\ A$. The rms value of the current is $12.5/\sqrt{2} = 8.84$ A. Hence the average power delivered to the resistor is

$$P = (8.84)^2 50 = 3906.25\ W$$

End of the Lesson (08.01.24)

Any questions?

New supplementary videos are added to the playlist on Youtube:

Supplementary videos: Enter Youtube-> Search for ‘Alparslan Zehir’ -> Go to Playlists -> Watch ‘Electrical Circuits’ playlist